CAD-bench: CAD Agent Reliability Signals

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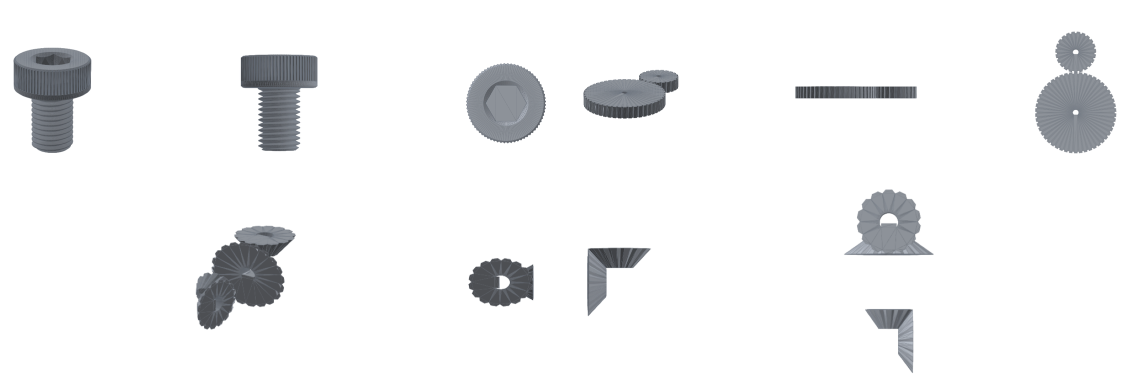
Agent Setting

Tool-using CAD agents do more than emit one file. They can:

- write CAD code and execute it;
- inspect exported geometry;
- revise errors across tool calls;
- submit a final STEP artifact.

For agents in open toolchains, reliability means the final artifact satisfies task-level mechanical requirements after interaction, not just that the workflow produced a render or a file.

Artifact Interface



Agents submit CAD code or final geometry. The scorer checks the resulting object, not the transcript.

Tasks	17 executable CAD requirements
Agent output	final CAD code or exported geometry artifact
Scorer input	executed artifact plus task contract
Runs	32 complete agent runs and 19 standalone baselines

Tool-Use Signals

Build	whether the toolchain produced an artifact
Repair	whether execution feedback improved static geometry
Interface	whether holes, sockets, threads, and joints are present
Behavior	whether assemblies transmit required motion

Construction

Mechanical task prompt with expected interfaces

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Agent writes, runs, inspects, and revises CAD artifacts

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Final submitted artifact after tool interaction

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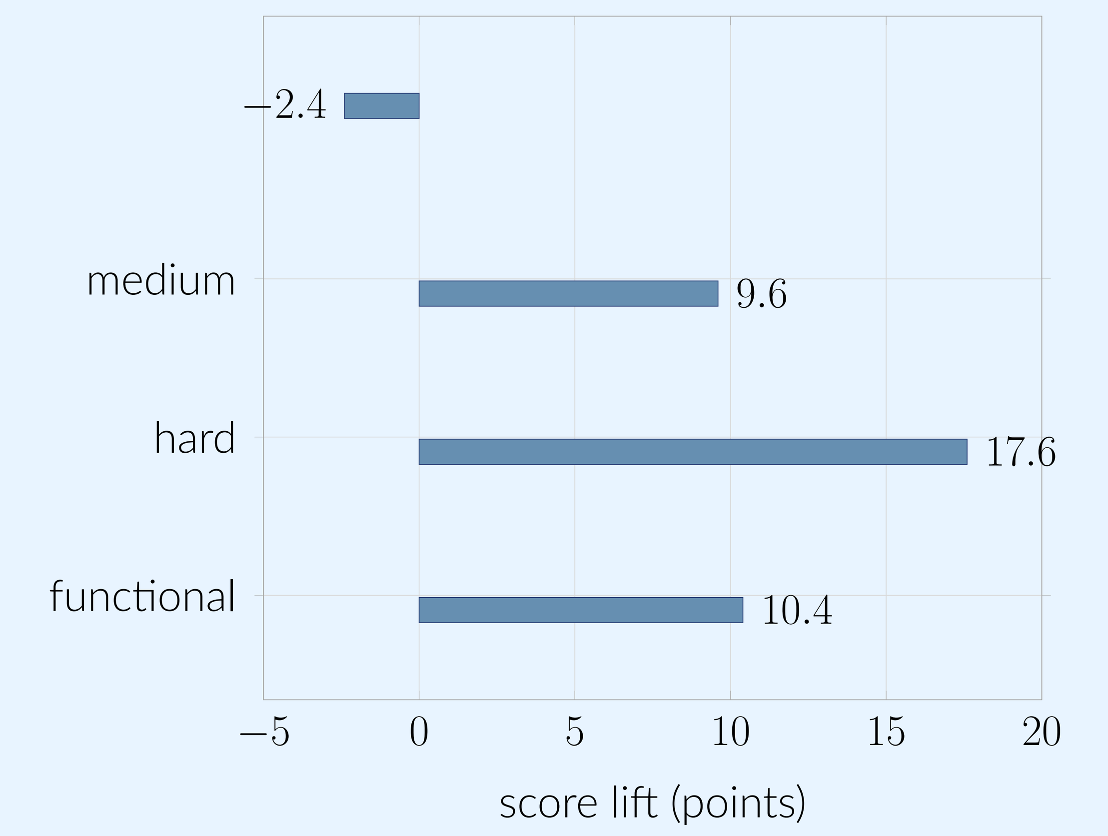
Fixed scorer checks geometry, interfaces, and function

One evaluator scores one-shot outputs and tool-using agent artifacts on the same tasks.

The artifact interface separates agent workflow quality from final mechanical correctness.

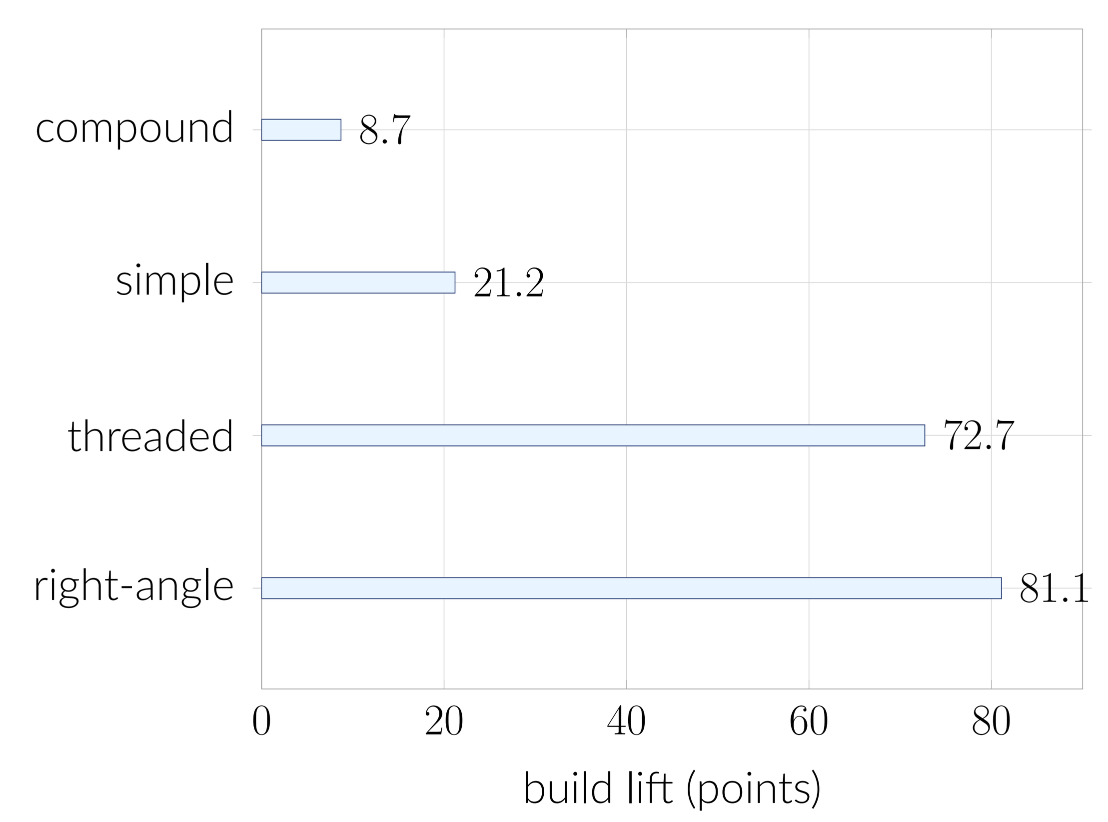
Contracts retain dimensions, joints, motion, and per-check traces before weighting.

Agent Lift



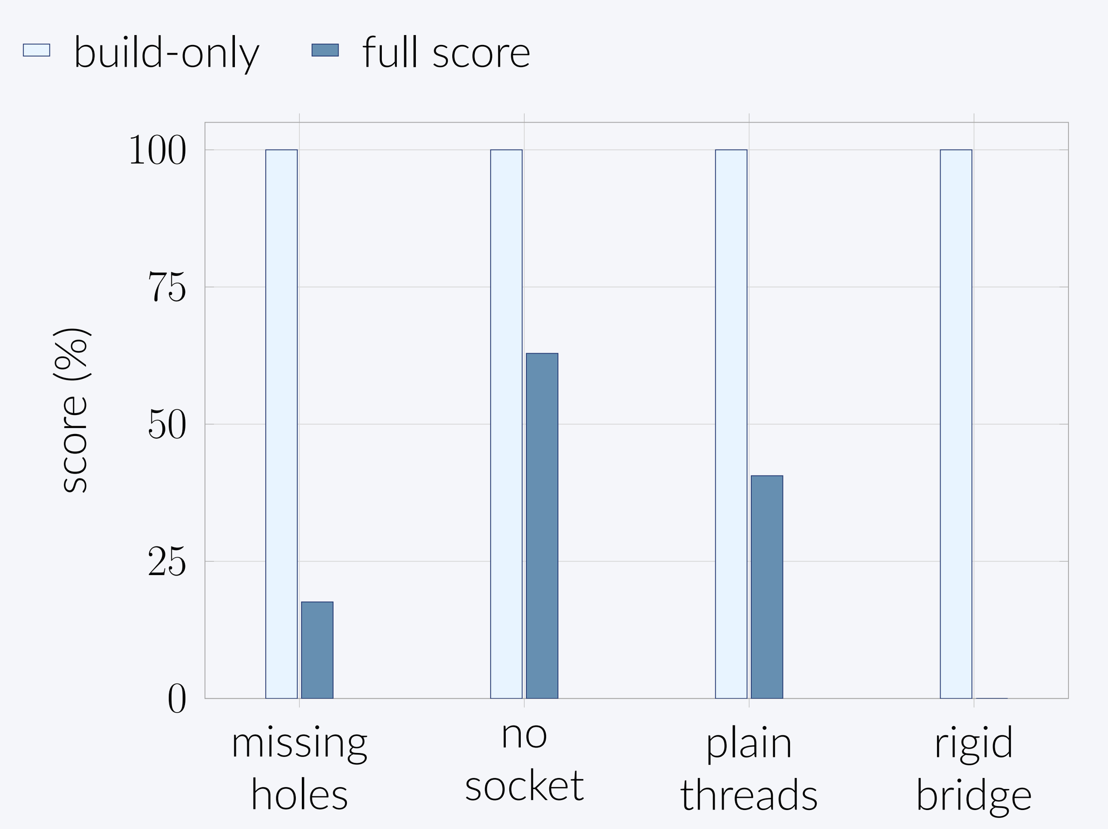
Across complete runs, agent harnesses improve hard static and functional categories, while easy tasks are already near saturation.

Tool-Use Build Lift



Tool use sharply increases build rates on functional tasks, especially the right-angle gearbox and threaded pair.

Behavioral Checks



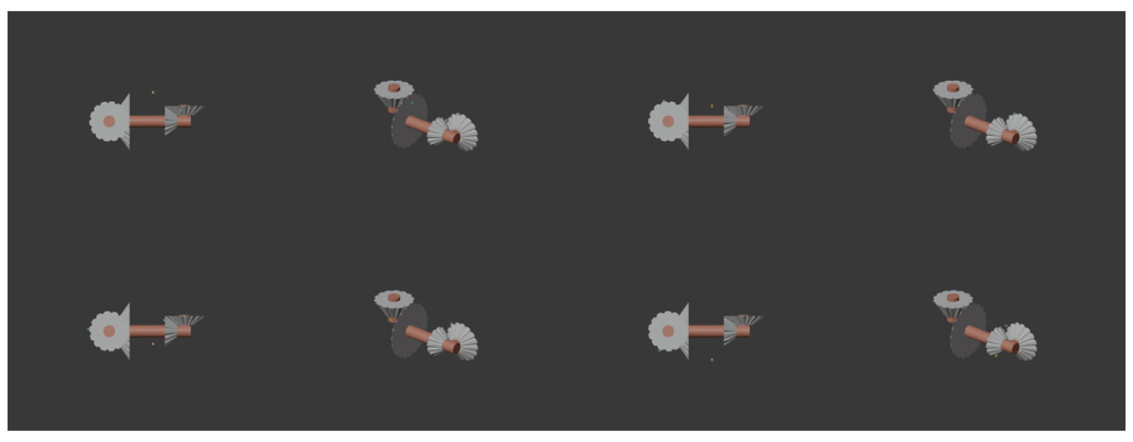
The same scorer records whether holes, sockets, mating threads, and motion behavior are actually present.

Agent Results

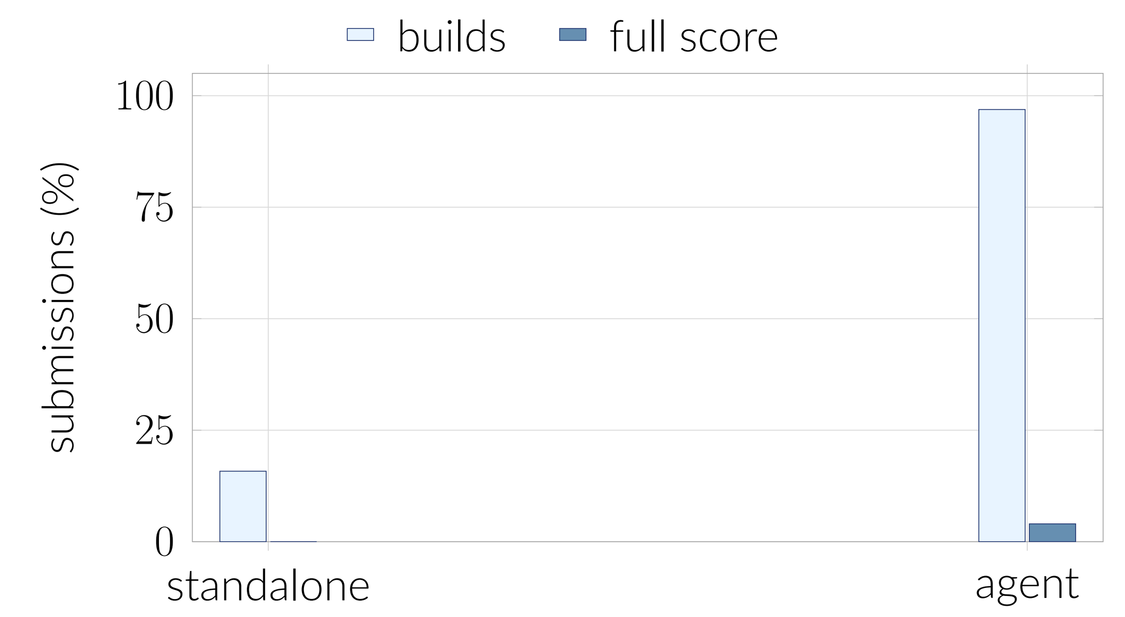
Best agent	67.7% overall, with strongest gains on hard static tasks
Functional avg	agent runs average 14.1% on functional mechanisms
Behavior split	scores separate tool progress from mechanical success
Standalone baseline	standalone runs provide a non-agent baseline for the same task set

Build, detail, and function scores remain separate before category weighting.

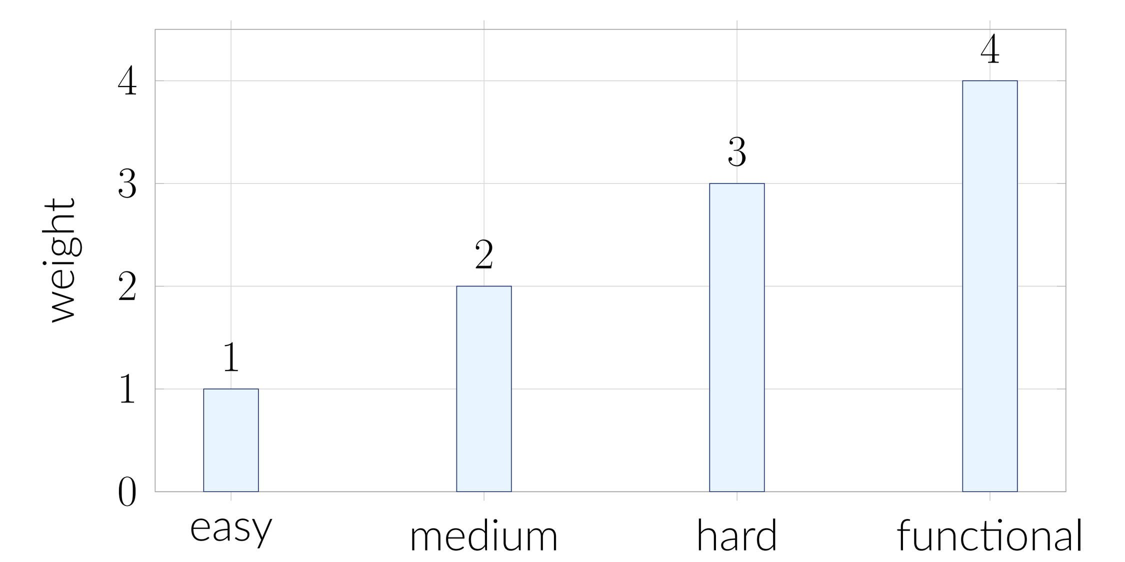
Right-Angle Gearbox



Reference motion frames for the right-angle gearbox. CAD Bench checks the required direction change, not just axle placement.

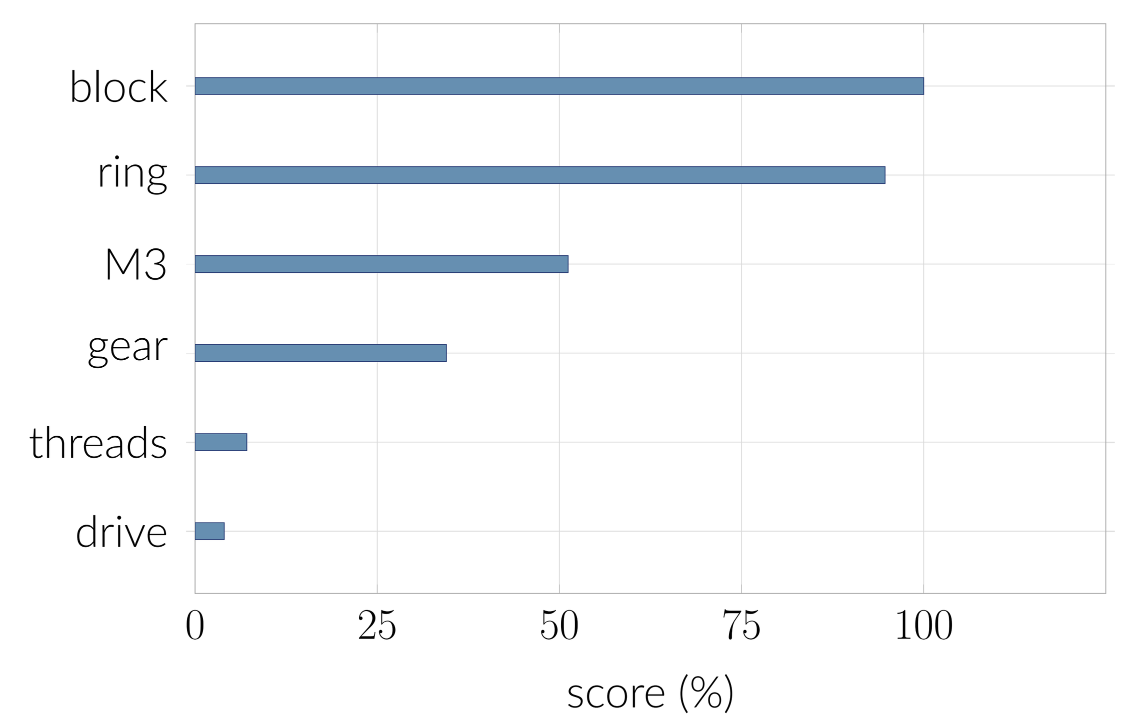


Score Weighting



The final score keeps functional behavior visible instead of letting the 13 static tasks dominate the four functional tasks.

Score Details



Agent scores remain high on simple static placements and drop where gear teeth, screw interfaces, mating threads, and motion matter.

Agent-System Evaluation

Tool-using agents can execute CAD programs, inspect generated artifacts, and revise outputs. CAD-bench scores the final object with fixed checks.

The score exposes where tool use improves construction and where interfaces or motion still require stronger planning.

That split keeps agent progress visible while preserving mechanical correctness.